MULTIMODAL AI · CHEST DISEASE SCREENING

AI-Assisted Multimodal Screening of Chest Diseases

Combining clinical information with chest X-ray images



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Problem Statement

"Preliminary disease assessment relies heavily on individual clinical evaluation, while symptoms and medical images are not always analysed together."

Need: an AI-assisted multimodal system that fuses clinical information with chest X-rays to screen selected chest diseases.



Depends on the individual

Screening quality varies with clinician experience and workload.



Data analysed in silos

Symptoms/history and X-ray findings are rarely interpreted together.



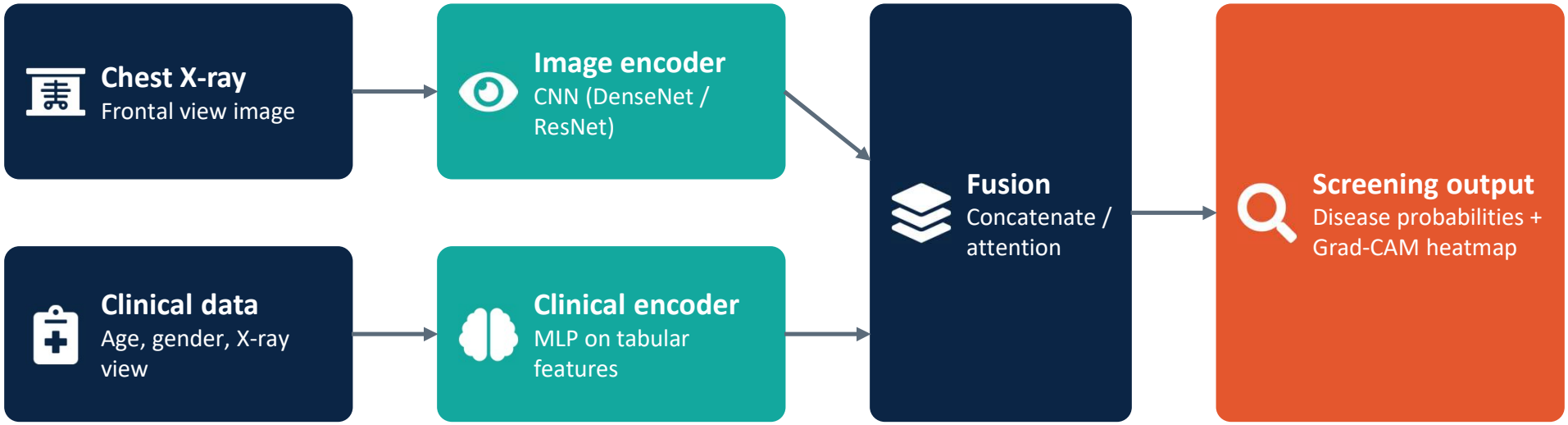
Delays in early screening

Manual reading of every image is slow, especially where radiologists are scarce.

Existing Approaches & Their Limitations

EXISTING APPROACH		LIMITATION
Manual radiologist reading of chest X-rays	→	Time-consuming; inter-observer variability; radiologist shortage
Clinical evaluation of symptoms alone	→	Subjective; cough, fever, breathlessness overlap across many diseases
Image-only deep learning (e.g., CheXNet, DenseNet CNNs)	→	Ignores patient context; more false positives; "black-box" output
Symptoms and images reviewed separately	→	No joint reasoning; fragmented information; slower decisions

Proposed System: Multimodal Fusion



Richer context
Demographics help disambiguate look-alike X-ray patterns

Explainable
Heatmaps show where the model looked

Screening aid
Supports, not replaces, the clinician

Methodology Workflow

1

Collect & preprocess

Resize, normalise, augment X-rays; encode and impute clinical features

2

Extract features

CNN backbone for images; MLP for clinical variables

3

Fuse modalities

Concatenate or attention-weight both feature vectors

4

Train & validate

Multi-label loss; patient-wise train/val/test split

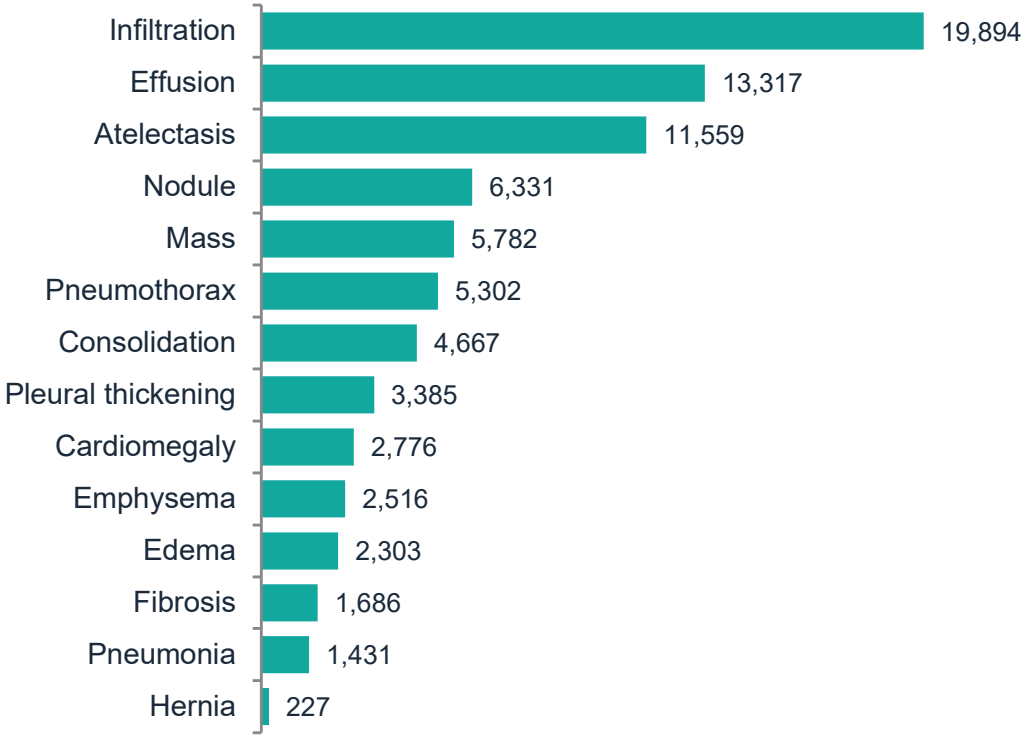
5

Explain & output

Probabilities per disease with Grad-CAM localisation

Dataset Used

NIH ChestX-ray14 — image counts per label (an image can carry several labels)



112,120 frontal-view chest X-rays

30,805 unique patients

14 thoracic disease labels

Labels are text-mined from radiology reports. **Clinical inputs:** age, gender, and X-ray view position from metadata (following Orhan et al., 2026, extended with age).

Research Foundation (1/3)

Orhan, Ucan, Alhajj & Kaya — Diagnostics, 2026

Scalable Unimodal and Multimodal Deep Learning for Multi-Label Chest Disease Detection: A Comparative Analysis

What it's about

Trained 12 models (ResNet50, EfficientNetB3, DenseNet121 × image-only vs. image+clinical-metadata) on the NIH ChestX-ray dataset at two scales (5,606 and 121,120 samples), evaluated by per-label AUROC. Multimodal fusion beat image-only consistently.

Limitations

Clinical metadata was limited to structured fields already in the NIH CSV (no real symptom text). Single dataset, single institution source — no external hospital validation. Simple fusion design; no attention-based or learned weighting between modalities tested.

What can be improved

Add richer clinical features (e.g. age, as we are doing); test intermediate/attention-based fusion instead of simple concatenation; validate on an external dataset (e.g. CheXpert) to check generalization.

Research Foundation (2/3)

Gapp, Tappeiner, Welk & Schubert — arXiv, 2024

Multimodal Medical Disease Classification with LLaMA II

What it's about

Used a 7-billion-parameter LLaMA II as the backbone for both text and vision, fine-tuned with LoRA, on the Open-I chest X-ray + report dataset. Compared early ("parallel"), late ("serial"), and mixed fusion by where the cross-attention layer sits. Early fusion won: 97.10% mean AUC, beating the prior TransCheX benchmark (96.29%).

Limitations

Very small test split (only 377 image-report pairs) for a 7B-parameter model — high overfitting risk. Single dataset (Open-I), no external validation. Per-class results were unstable: the Pneumothorax AUC dropped sharply in the serial/mixed pipelines (as low as 0.774–0.840), and overall gains over TransCheX were marginal (<1% in several configurations).

What can be improved

Test on a larger multimodal dataset (e.g. MIMIC-CXR) to see if gains hold at scale; add interpretability analysis (the authors flag this as future work); investigate why specific classes like Pneumothorax are unstable across fusion strategies.

Research Foundation (3/3)

Gundapaneni, Zhi & Rodrigues — UCL, arXiv, 2024

Deep Learning-Based Noninvasive Screening of Type 2 Diabetes with Chest X-ray Images and Electronic Health Records

What it's about	Built a fused dataset from MIMIC-IV (EHR) + MIMIC-CXR (X-rays) + MIMIC-IV-ECG, then compared an early-fusion ViLT transformer against a joint-fusion ResNet+LSTM. The joint ResNet-LSTM model won: 0.86 AUROC, a 2.3% improvement over an X-ray-only baseline, using only 9,863 training samples. Ran rigorous ablations: removing pre-training, adding noise, and simulating missing X-rays.
Limitations	The authors state their own key limitation directly: repeated ICU stays from the same patient were treated as separate samples despite sharing the same X-ray/ECG, which may bias the model toward chronic conditions. No external-hospital validation — all data is from one medical centre (BIDMC). The ViLT transformer overfit badly on this small, out-of-domain dataset and was far less noise-robust than the CNN-based model.
What can be improved	External validation on a second hospital's data, as the authors explicitly call for; de-duplicate or weight repeated patient visits to reduce bias; explore pre-training ViLT-style transformers on in-domain medical data instead of general-purpose weights.

Evaluation Plan & Expected Outcomes

AUROC
Per-disease discrimination

Sensitivity
Catch true disease cases

Specificity
Limit false alarms

F1-score
Precision–recall balance

Ablation design

- Image-only baseline (CNN)
- Clinical-only baseline (MLP)
- Proposed multimodal fusion model
- Compare with paired significance tests

Expected outcomes

- Fusion outperforms either modality alone
- Fewer false positives on look-alike findings
- Heatmaps that align with radiologist attention

Applications, Challenges & Ethics

Applications

- Primary-care and rural triage
- Second-reader support for radiologists
- Prioritising urgent cases in worklists
- Large-scale screening programmes

Challenges & Ethics

- Dataset bias and label noise
- Generalisation across hospitals/devices
- Patient privacy and data consent
- A screening aid, not a diagnosis

Conclusion & Future Scope

- Combining symptoms with chest X-rays gives a more complete screening signal than either alone.
- Fusion + Grad-CAM offers accuracy and explainability for selected chest diseases.
- Future: clinical notes via NLP, multi-hospital validation, lightweight deployment for low-resource clinics.



Thank you · Questions?